

ALTUS QUARTZ MT, OK, USA

WMO: 720354

Lat: 34.699N

Lon: 99.338W

Elev: 1433

StdP: 13.95

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 63901

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.0	21.1	2.8	6.7	24.4	7.8	8.6	28.6	32.9	44.4	28.7	45.9	9.4	10	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	104.2	70.4	100.8	71.0	98.6	71.1	76.8	91.6	75.5	90.5	74.7	89.3	12.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	128.6	82.3	72.0	124.9	81.4	70.8	119.6	80.7	41.2	91.3	40.1	90.5	39.1	89.2	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
28.1	25.1	22.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			9.5	105.9	4.8	3.9	6.1	108.6	3.2	110.9	0.5	113.1	-3.0	115.9		
			7.4	78.0	4.7	1.8	4.0	79.3	1.3	80.4	-1.4	81.4	-4.8	82.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.5	41.4	45.1	55.2	62.4	72.3	82.4	84.5	83.6	75.8	63.9	52.2	41.9
	DBStd	17.66	9.20	11.01	10.15	9.10	8.19	5.38	4.98	5.14	6.30	9.13	9.13	8.83
	HDD50	937	293	200	65	15	0	0	0	0	0	10	80	274
	HDD65	3043	732	559	329	151	29	0	0	0	5	133	390	715
	CDD50	5860	25	64	224	387	691	972	1070	1040	775	441	147	24
	CDD65	2491	0	3	23	73	254	522	605	575	330	99	7	0
	CDH74	31158	21	83	335	971	2892	6718	8220	7323	3362	1083	141	9
	CDH80	16789	2	20	87	379	1403	3848	4871	4204	1563	381	31	0
Wind	WSAvg	10.2	9.5	10.8	11.8	12.4	11.6	11.6	8.9	8.2	8.7	9.7	9.5	9.2
Precipitation	PrecAvg	27.0	1.0	1.1	1.8	2.7	3.7	3.7	2.6	2.6	2.3	2.4	1.7	1.2
	PrecMax	40.4	2.8	4.1	4.5	8.0	10.9	8.5	7.8	7.4	7.1	5.9	6.5	4.1
	PrecMin	16.8	0.0	0.0	0.0	0.1	0.2	0.0	0.2	0.0	0.2	0.0	0.0	0.0
	PrecStd	6.5	0.8	1.1	1.3	1.8	2.4	2.3	1.7	2.0	1.6	1.5	1.7	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.1	82.4	87.8	95.2	101.1	106.3	107.9	108.5	99.5	92.3	84.4	73.5
		MCWB	55.0	56.5	59.6	63.8	65.2	69.1	69.6	69.5	70.2	66.4	60.3	54.8
	2%	DB	69.7	75.2	81.7	89.3	96.9	102.0	104.0	103.6	95.0	88.4	78.2	67.2
		MCWB	50.2	55.0	60.2	63.6	68.3	70.4	71.3	70.5	70.8	66.7	59.2	52.7
	5%	DB	63.5	69.6	77.3	84.0	92.9	99.1	100.9	99.6	91.6	84.3	72.8	61.4
		MCWB	47.9	52.9	59.3	62.8	67.4	70.8	71.5	71.8	71.1	65.7	57.1	49.0
	10%	DB	58.4	63.8	72.5	79.3	88.3	96.2	98.1	96.7	89.1	80.6	68.1	56.8
		MCWB	45.5	50.5	57.1	61.7	67.1	71.3	71.8	72.3	70.5	63.3	55.7	47.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.4	62.1	67.1	70.1	74.7	77.4	78.3	78.4	76.6	73.4	65.4	61.1
		MCDB	70.1	70.8	78.4	82.5	87.0	92.8	91.8	92.5	89.3	84.2	73.8	66.8
	2%	WB	52.5	58.5	65.3	67.9	73.2	75.4	76.5	77.0	75.0	71.2	62.3	55.1
		MCDB	64.6	69.9	75.8	80.1	86.7	91.4	91.1	92.4	87.5	81.8	71.5	62.0
	5%	WB	49.0	55.1	62.8	66.3	71.8	74.4	75.3	75.5	73.6	68.9	60.0	50.9
		MCDB	61.7	67.1	71.2	77.9	84.9	90.2	90.4	90.6	85.0	78.5	69.0	60.1
	10%	WB	46.1	51.3	59.9	64.4	70.2	73.3	74.3	74.6	72.4	66.4	57.3	47.2
		MCDB	57.0	62.9	69.0	74.8	83.0	88.9	89.3	88.9	83.7	77.2	66.9	56.6
Mean Daily Temperature Range	5% DB	MDBR	25.1	25.3	25.6	25.4	23.8	23.5	23.0	22.6	22.0	24.9	25.3	21.7
		MCDBR	35.5	35.5	31.5	33.5	30.0	27.6	27.3	26.7	25.5	28.8	32.2	30.8
		MCWBR	19.6	18.5	15.4	13.8	9.7	6.8	6.0	7.4	8.3	11.7	16.2	17.9
	5% WB	MCDBR	32.5	31.5	24.5	26.0	23.8	23.3	22.2	21.9	20.7	23.1	25.6	26.6
		MCWBR	18.8	18.6	13.6	13.4	10.0	7.8	7.1	7.8	7.9	11.7	15.2	17.1
Clear-Sky Solar Irradiance	taub	0.282	0.297	0.331	0.364	0.385	0.412	0.415	0.405	0.373	0.342	0.311	0.284	
	taud	2.537	2.496	2.420	2.351	2.367	2.326	2.357	2.333	2.398	2.473	2.489	2.530	
	Ebn at Noon	297	303	299	292	285	277	275	276	281	283	282	289	
	Edh at Noon	25	30	35	39	39	41	39	39	35	30	26	24	
All-Sky Solar Radiation	RadAvg	965	1161	1544	1881	2032	2236	2227	2023	1690	1297	1004	820	
	RadStd	94	153	99	144	210	186	124	117	110	172	124	81	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page